

AI-generated fake review detection[☆]

Jiwei Luo^a, Guofang Nan^{a,*}, Dahui Li^b

^a International Business School, Hainan University, Haikou 570228, PR China

^b College of Business, University of Colorado Colorado Springs, CO 80918, USA

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ABSTRACT

Online reviews of e-commerce platforms have long been recognized as a major factor that influences a consumer's purchasing decisions. However, the emergence of generative artificial intelligence (GAI) has accelerated the proliferation of fake online reviews, which can significantly reduce consumer trust in these platforms. This study proposes a novel supervised learning approach that can be flexibly integrated into decision support system to help platforms effectively detect AI-generated fake reviews. In the approach, we first construct two types of variables to distinguish between human-written genuine reviews and AI-generated fake reviews. Then, we introduce an outlier detection method based on cumulative probability density to calculate the probability that a fake review generated by AI. Finally, we train the AdaBoost model using the cumulative probability density values of reviews computed above to obtain classifier that can accurately detect AI-generated fake reviews. Numerical experiments demonstrate that the proposed method can produce more accurate detections of AI-generated fake review than several state-of-the-art baseline methods. We contribute to the related literature by the exploitation of statistical theory, which posits that outliers, as small probability events, are typically located at the tails of feature distributions, a principle effectively employed in detecting AI-generated fake reviews.

1. Introduction

“Online product reviews are becoming a battlefield for modern AI”.

— Annie Probert/ NBC News [1]

Online reviews have continued to play a pivotal role in shaping consumer behavior on e-commerce platforms because consumers rely on these reviews when making purchasing decisions, which is demonstrated in prior studies [2–4]. Notably, nearly 90% of online shoppers report that authentic reviews influence their buying choices and that they place greater trust in peer feedback than in the product's price itself.¹ However, the review system is undermined by the unethical practices of certain merchants who employ tools to inflate product rankings and damage competitor reputations [5], which in turn erodes consumer trust and compromises the integrity and efficiency of the review system [6].

To address this growing concern, platform companies have deployed decision support systems to detect and filter out fraudulent reviews [7, 8]. These technological interventions aim not only to curb the dissemination of fake content but also to mitigate its broader negative impact on the platform ecosystem [9,10]. By reinforcing the authenticity of user-generated content, these systems play a critical role in preserving the credibility and functionality of digital marketplaces.

In recent years, the emergence of GAI has accelerated the spread of fake reviews across digital platforms, driven by merchants who can leverage GAI to rapidly generate large volumes of fake reviews with minimal effort.^{2,3} Unlike traditional fake reviews, those produced by GAI are often more persuasive and harder to detect. This is due to GAI's advanced capability to produce coherent, contextually relevant, and human-like language, which closely mimics the tone and structure of authentic user feedback [11,12]. As a result, these GAI-crafted reviews pose a greater threat to consumer trust and platform integrity. The superior linguistic features offered by GAI have attracted more malicious actors to apply GAI to generate deceptive content, thereby intensifying the proliferation of fake reviews on e-commerce platforms.

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* Corresponding author.

E-mail addresses: jiweiluo@hainanu.edu.cn (J. Luo), gfnan@hainanu.edu.cn (G. Nan), dli@uccs.edu (D. Li).

¹ Please see <https://emplifi.io/press/study-reveals-customer-ratings-reviews-impact-on-purchase-decisions> (accessed on February 28, 2023).

² Please see <https://searchengineland.com/new-google-policy-says-ai-generated-reviews-are-spam-and-against-merchant-center-policies-429984> (accessed on July 28, 2023).

³ Please see <https://www.cnbc.com/2023/04/25/amazon-reviews-are-being-written-by-ai-chatbots.html> (accessed on April 25, 2023).

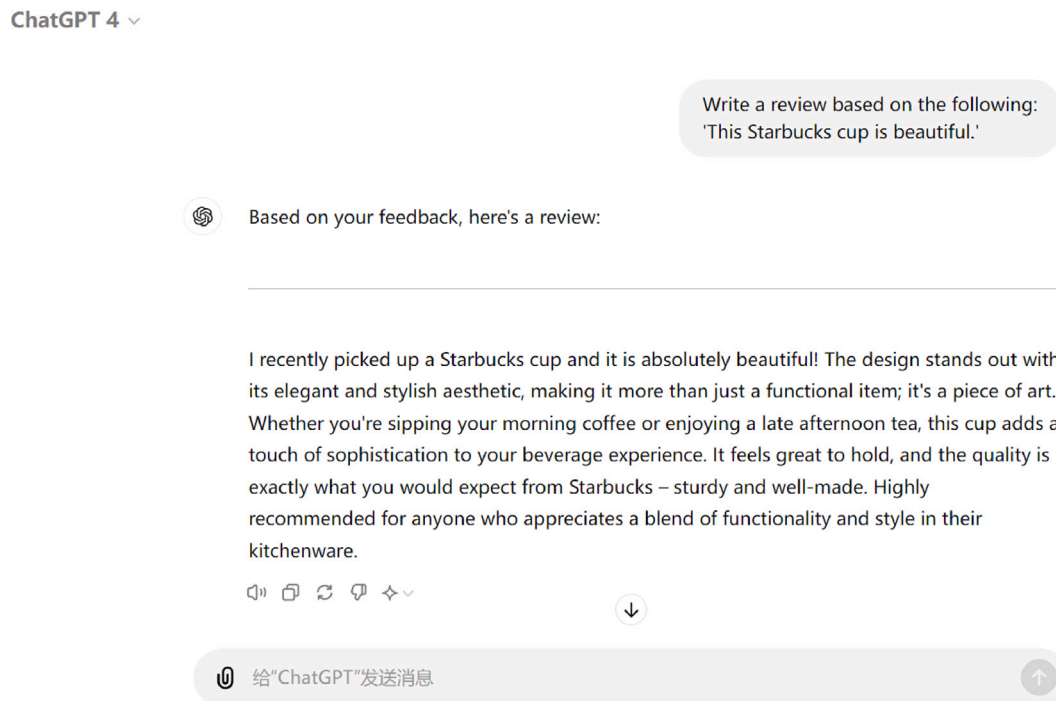


Fig. 1. Example of an AI-generated fake review.

At the same time, regular consumers who genuinely intend to share their experiences and endorse products or merchants are also turning to AI tools to generate their reviews. While this may streamline the review-writing process, it introduces a critical limitation: AI lacks the capacity to fully comprehend the nuanced, real-life experiences of individual consumers.⁴ As a result, AI-generated reviews may exhibit a higher degree of bias compared to those written by humans. Fig. 1 illustrates a review generated with the assistance of ChatGPT (note: product details are fictitious). This inherent bias in AI-generated content can undermine consumer trust in both the product and the associated merchant.³ Given these challenges, it is imperative to propose a novel detection method specifically designed to identify AI-generated fake reviews. Such a system enable platforms to swiftly flag or filter out inauthentic content, thereby preserving the integrity of user feedback and maintaining trust within the platform ecosystem.

Although AI-generated fake reviews may appear indistinguishable from those written by humans, they often exhibit subtle yet identifiable linguistic patterns indicative of their artificial origin [13]. Notably, these reviews frequently replicate stylistic markers commonly found in traditional deceptive content used by businesses. For example, counterfeit reviews tend to feature exaggerated emotional expressions, as their primary objective is to either excessively praise or unjustly criticize a product's reputation [14]. Similarly, when merchants employ AI tools, the reviews often display heightened emotional intensity. This pattern underscores the importance of incorporating conventional linguistic cues into detection strategies. By analyzing these textual characteristics, platforms can more effectively identify and flag AI-generated fake reviews, even when these reviews closely mimic authentic user feedback.

In addition, the lexical choices in AI-generated reviews are typically determined by selecting words with the highest predicted probability, as computed by the underlying Large Language Model (LLM). As a result, the textual output of such reviews tends to exhibit lower

randomness compared to content authored by humans. This reduction in linguistic variability can be quantitatively assessed using two advanced linguistic metrics: perplexity and burstiness [15,16]. These measures evaluate the predictability and distributional irregularities in text, offering insight into the generative behavior of LLMs. Given their diagnostic potential, it is essential to incorporate these novel linguistic features into the detection framework for AI-generated fake reviews. Doing so can enhance the precision of detection systems by identifying subtle statistical patterns that distinguish synthetic content from authentic human expression.

In addition to linguistic-based approaches, automation techniques, particularly those leveraging deep learning algorithms, have gained momentum in the detection of AI-generated content. A notable example is OpenAI, which has fine-tuned LLMs to develop a detector that can identify up to 95% of texts generated by GPT-2.⁵ Despite this promising performance, such models face two critical limitations: a lack of interpretability and reduced effectiveness when applied to short texts such as online reviews. These constraints hinder their practical deployment in real-world practice.

Prior research has explored outlier detection methods by modeling atypical reviewer behavior as a means of identifying counterfeit reviews [6,9]. For instance, Kumar et al. proposed a technique that constructs optimal sub-distributions representing various reviewer behavioral traits, which are then aggregated into a composite distribution to detect anomalous reviewers [17]. Similarly, Li et al. introduced an approach based on cumulative probability density, identifying outliers as rare events located in the tails of the distribution [18]. However, a persistent challenge in these methods lies in determining an appropriate probability threshold to distinguish outliers from normal behavior. Without a clear boundary, the accuracy of these models remains constrained. Therefore, it is imperative to develop an innovative outlier detection methodology that overcomes these limitations, particularly by enhancing threshold calibration and interpretability, to enable more precise identification of AI-generated fake reviews.

⁴ Please see <https://seo.ai/blog/ai-generated-reviews-are-spam> (accessed on August 4, 2023).

⁵ Please see <https://openai.com/research/gpt-2-1-5b-release>.

To address these limitations, we propose a multi-faceted methodological framework. First, we construct a comprehensive set of variables that integrates both traditional linguistic features and novel linguistic indicators. This enriched feature set enables a more nuanced analysis of review authenticity. Next, we introduce an innovative technique based on cumulative probability density (CPD) to estimate the likelihood that a given review is generated by AI. This probabilistic measure serves as a foundational input for the subsequent classification phase. Finally, we utilize the computed CPD values to train a suite of established classification models. These models are then used to develop classifiers capable of accurately identifying AI-generated fake reviews. To the best of our knowledge, this study represents one of the first attempts to develop an interpretable machine learning approach specifically tailored to the detection of AI-generated fake reviews. By offering a transparent and effective detection mechanism, our approach provides critical decision support for platform operators to flag inauthentic content. This contribution is essential for the reputation of e-commerce platforms.

The remainder of this study is organized as follows. First, we review the existing literature on methods for detecting AI-generated fake reviews. Second, we introduce the two categories of variables constructed in this study—traditional linguistic features and novel linguistic characteristics. Third, we present our proposed anomaly detection method based on cumulative probability density. Next, we evaluate the performance of the proposed method using three real-world datasets. Finally, we offer a discussion of the findings and conclude with implications and future research directions.

2. Literature review

2.1. Fake review detection

Fake reviews on e-commerce platforms refer to deliberately misleading evaluations crafted by individuals with the intent to manipulate product perceptions or inflate ratings [14,19]. These reviews often exhibit identifiable linguistic and behavioral patterns, such as exaggerated sentiment, unnatural phrasing, or inconsistencies in reviewer history [10,17,20]. Researchers have leveraged these suspicious cues to develop a variety of fake review detection methods aimed at distinguishing genuine feedback from deceptive content [8].

More recently, the rapid advancement of LLMs has introduced new challenges. While LLMs have demonstrated impressive capabilities in learning from vast datasets [21], their application, particularly in commercial contexts, has raised ethical concerns, including the potential for misuse in generating fake content. For example, TripAdvisor, the world's largest travel platform, reported deleting over 20,000 AI-generated fake reviews following the initial release of ChatGPT.⁶ Compared to human-written fake reviews, those generated by AI tend to be grammatically correct and logically coherent, making them more difficult to detect using traditional methods. To address this emerging threat, recent studies have proposed detection techniques based on transfer learning and deep learning.⁷ These approaches aim to adapt to the evolving nature of AI-generated content. The next section reviews recent research on the input features and corresponding detection methods used to identify both human-written and AI-generated fake reviews.

⁶ Please see <https://www.theguardian.com/money/2023/jul/15/fake-reviews-ai-artificial-intelligence-hotels-restaurants-products> (accessed on July 15, 2023).

⁷ Please see <https://openai.com/blog/new-ai-classifier-for-indicating-ai-written-text>.

2.2. Fake reviews detection methods

2.2.1. Human-written fake reviews detection methods

Detecting fake reviews involves leveraging predefined features to distinguish deceptive content from authentic user feedback [10,22]. Prior studies have categorized these features into three types: content features, behavioral features, and relationship features [6]. Content features such as sentiment polarity, lexical diversity, and readability capture linguistic irregularities that characterize fabricated reviews [6,7]. Behavioral features, including posting frequency and temporal patterns, help identify suspicious reviewer activity [17]. Relationship features, such as reviewer-product interaction histories, review similarity networks, and social graph structures, enhance detection by modeling the relational dynamics among users and content [6]. Building on these features, machine learning-based approaches have been proposed to detect fake reviews [8,23,24]. For example, Yin et al. employed supervised learning techniques using linguistic features like sentiment polarity [25]. Similarly, Kumar et al. developed a probabilistic model that integrates both linguistic and behavioral traits of reviewers [7]. Another line of research has explored the use of word embedding techniques such as Word2Vec and Doc2Vec to map textual reviews into dense vector representations. These embeddings capture subtle semantic and syntactic differences between genuine and deceptive content. By transforming reviews into high-dimensional semantic spaces, deep learning models can extract discriminative features that often outperform traditional machine learning models in detecting nuanced patterns [8,26,27]. However, despite their predictive power, deep learning models suffer from limited interpretability. Due to the complexity of their architectures that comprise numerous neurons and nonlinear transformations, it remains challenging to intuitively understand the role of individual parameters or neurons [28]. Even when the full network structure and parameter values are known, the models often function as “black boxes” [29].

2.2.2. AI-generated fake reviews detection methods

Although AI-generated reviews may be difficult for humans to recognize, this does not imply that such reviews are devoid of suspicious characteristics. Prior research has shown that LLMs exhibit distinctive linguistic patterns that can be leveraged to differentiate AI-generated fake reviews from human-authored content. LLMs generate text by selecting words with the highest predicted probability from their vocabulary, resulting in output that is highly legible but often exhibits templated sentence structures. In contrast, human-written reviews tend to display greater lexical diversity and unpredictability [30]. To quantify these differences, researchers have developed various metrics to evaluate LLM performance in the context of fake review detection [15,31]. For example, GPTZero, a widely recognized detection tool, uses two key metrics, perplexity and burstiness, to measure textual randomness.

In parallel, others have explored transfer learning-based approaches. For instance, Salminen et al. fine-tuned a RoBERTa model and demonstrated its effectiveness in identifying GPT-2-generated fake reviews [32]. Gambetti et al. proposed a fine-tuned GPT-Neo model that achieved high accuracy in distinguishing AI-generated reviews from human-written ones [33]. However, a major limitation of these deep learning-based methods is the lack of interpretability and transparency, which poses challenges for the practical deployment.

It is also worth noting that while some studies have utilized LLM evaluation metrics as novel linguistic features for detecting AI-generated fake reviews, none have incorporated traditional linguistic features commonly used in detecting human-written fake reviews. This is an oversight, as AI-generated reviews, particularly those crafted by merchants using AI tools, often exhibit suspicious cues similar to those found in deceptive human-authored content. To address this gap, our study integrates both types of features, i.e., novel linguistic indicators derived from LLM behavior and traditional linguistic features used in fake review detection, into a unified detection framework. This

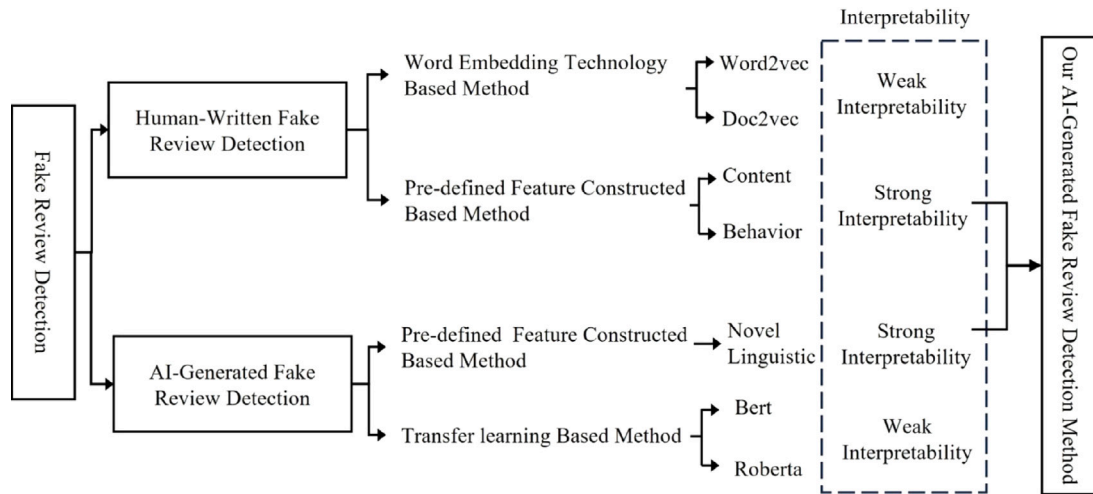


Fig. 2. Fake reviews detection methods.

Table 1
An overview of ADSAEO.

Source	Description
Kumar et al. [17]	Based Mixture Model Learning
Luo et al. [6]	Based General Mixed Probability Method
Kumar et al. [7]	Based cumulative effects of features distributions
Kumar et al. [9]	Based Hierarchical Supervised Learning
Li et al. [18]	Based Empirical Cumulative

hybrid approach enables the construction of a more comprehensive set of suspicious cues for identifying AI-generated fake reviews. Fig. 2 illustrates the connections and distinctions between existing fake review detection methods and our proposed approach.

2.3. Anomaly detection methods

AI-generated fake reviews often differ significantly from genuine human-authored reviews in terms of feature distributions, making these disparities a valuable basis for detection [15,34]. Accordingly, it is essential to develop an innovative outlier detection methodology that capitalizes on these distributional differences to effectively identify AI-generated fake reviews. A substantial body of research has explored statistical anomalies in reviewer behavior as a means of detecting fraudulent content [7]. For example, Luo et al. proposed a comprehensive mixed probability model that identifies fake reviews by analyzing discrepancies in feature distributions between authentic and deceptive content. Their numerical experiments demonstrated that this approach outperformed several baseline methods in accurately detecting fraudulent reviews [6].

To support this line of inquiry, Table 1 presents a catalog of anomaly detection methods that leverage statistical irregularities exhibited by outliers, referred to here as Anomaly Detection via Statistical Anomalies Exhibited by Outliers (ADSAEO). These methods collectively underscore the potential of statistical modeling in enhancing the precision of fake review detection systems.

However, most of the techniques listed in Table 1 do not strictly adhere to the classical statistical principle that defines outliers as data points located at the extremities of feature distributions. Among these methods, only the approach proposed by Li et al. explicitly incorporates this principle by identifying outliers as rare events situated in the tails of skewed distributions [18]. Nevertheless, their method faces a significant challenge in determining an appropriate cumulative probability threshold to effectively distinguish outliers from normal observations, an issue that increases the risk of misclassification. This limitation

Table 2
An overview of other types of anomaly detection methods.

Source	Description	Application
Van et al. [35]	Based Representation Learning	Credit card fraud detection
Yu et al. [36]	Based Graph Mining	Network intrusion detection
Liu et al. [37]	Single-Objective Generative Adversarial Active Learning	General method
Yu et al. [38]	Based convolutional neural network	Fall detection
Chakraborty et al. [39]	Based ensemble of probabilistic neural networks	General method

underscores the need for a novel outlier detection methodology that can overcome the shortcomings of existing approaches. Such a method should not only align with foundational statistical theory but also offer practical mechanisms for threshold calibration to enhance detection accuracy.

In addition, prior research has introduced various outlier detection techniques applicable across domains such as credit card fraud detection. For example, Van et al. developed a representation-learning-based method that models transactional relationships between cardholders and merchants to identify anomalous behavior [35]. Table 2 summarizes several types of anomaly detection methods that have been successfully applied in diverse application areas. However, a common drawback of these methods is their reliance on a large number of hyperparameters, which compromises their interpretability. This complexity makes it difficult for practitioners to understand or explain the model's decision-making process, thereby limiting their practical utility in high-stakes environments such as e-commerce platforms.

3. Research framework

In this section, we introduce an AI-generated fake review detection method termed AGFRDCP (AI-Generated Fake Review Detection via Cumulative Probability), as illustrated in Fig. 3. The overall framework comprises three key stages.

First, we construct two distinct categories of variables, including the traditional and novel linguistic features, as discussed in earlier sections. These variables are then transformed into optimal univariate distributions, allowing for statistical differentiation between AI-generated fake reviews and genuine human-authored reviews [6,9]. Second, we propose a Cumulative Probability-based Method (CPM). For each review,

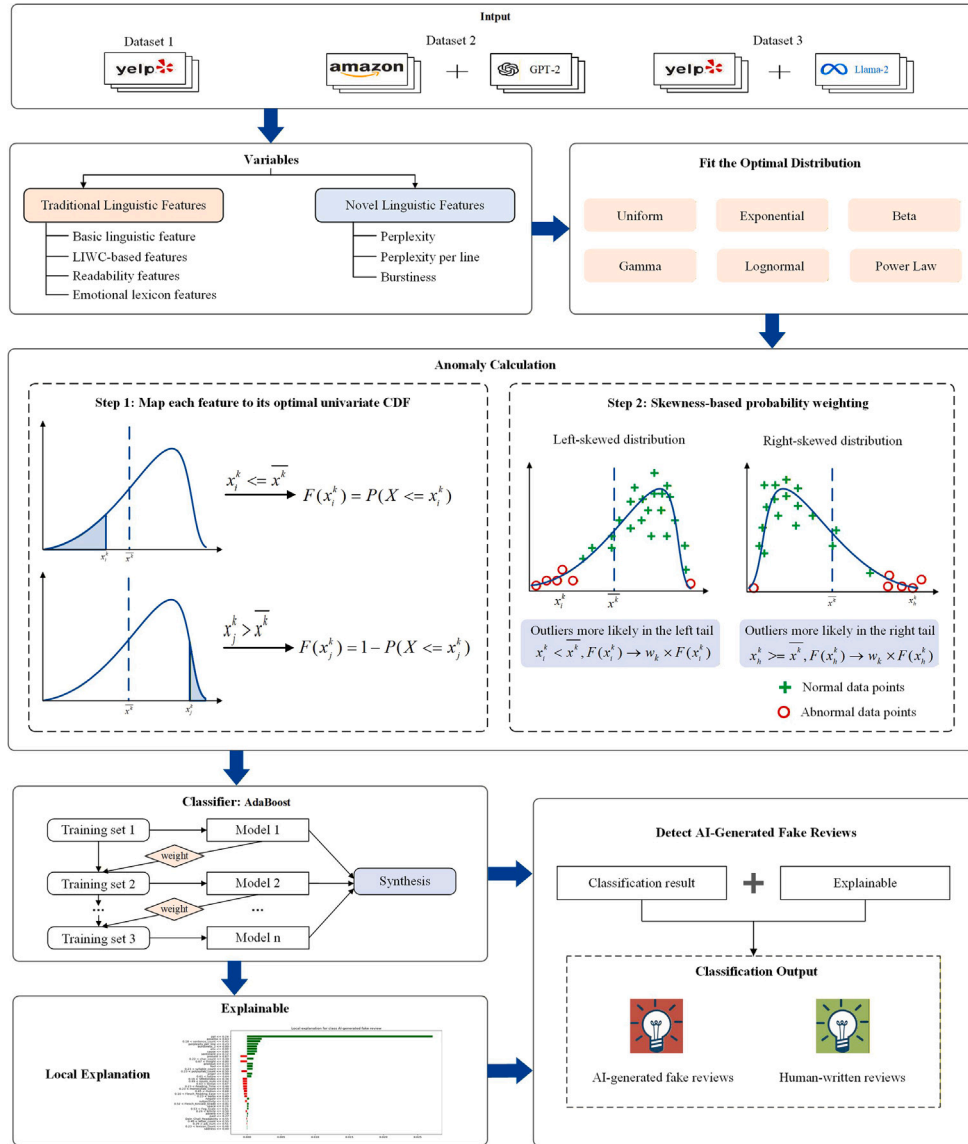


Fig. 3. The overall design of AGFRDCP.

we compute the cumulative probability that its variables conform to their respective distributions. Given that most variable distributions are skewed, we anticipate that AI-generated fake reviews are more likely to appear on one side of the distribution. To account for this, we apply a weighting scheme to the cumulative probabilities based on the skewness of each distribution. Finally, we use the weighted cumulative probability values as input features to train an AdaBoost classification model. This model is then evaluated using multiple real-world datasets to assess its effectiveness in detecting AI-generated fake reviews.

Overall, consistent with prior research on detection methodologies in the field of information systems [17], the foundational assumption of our proposed method is that while AI-generated fake reviews may evade detection when evaluated through isolated linguistic features, their systematic deviations from organic human writing patterns will manifest as statistical anomalies, particularly in the tail ends of multiple feature distributions. These deviations provide a robust basis for distinguishing AI-generated content from authentic human reviews. The remainder of this section details the design and implementation of the proposed AGFRDCP framework.

3.1. Traditional linguistic features

AI-generated fake reviews may appear indistinguishable from those written by humans. However, when AI is used by merchants as a tool to manipulate reviews, certain traditional linguistic features can still reveal suspicious cues. For example, as shown in Table 3, three fake reviews generated by GPT-3 do not significantly differ from genuine human-written reviews in terms of grammar or sentence structure. Nevertheless, these reviews exhibit strong emotional tendencies, which can serve as red flags for detection.

To capture such patterns, we construct four categories of traditional linguistic features, i.e., basic linguistic features, Linguistic Inquiry and Word Count (LIWC)-based features, readability features, emotional lexicon features.

3.1.1. Basic linguistic feature

Basic linguistic features of review texts play a critical role in detecting AI-generated fake reviews. For example, merchants may use AI tools to craft reviews that include an excessive number of adjectives or adverbs to exaggerate product or service experiences. These stylistic

Table 3
Example of AI-generated fake reviews.

	Text
1	I visited Seamores for dinner with my wife and we both had a great time. The atmosphere is very elegant and bright, perfect for a spring or summer date. The fresh haddock was cooked perfectly and the sides were delicious. I would highly recommend the Reel Deal to anyone coming here.
2	I always enjoy a meal at Sushi-Ko. The food is always fresh and the service is always friendly and reliable. The atmosphere is always fun and the menu features a variety of interesting and unique Japanese dishes. I've taken my mom, friends, big groups, and small groups here, and everyone has had a great time. The music is always good and the seasonal menu items are always tasty. I would definitely recommend making a reservation so that you can enjoy a meal with a group. Sushi-Ko is always busy, so it's a great place to go for a night out.
3	I had a generally negative experience at this restaurant. The pasta was overpriced and the drink was watered down, which made the overall meal not very enjoyable. The ambiance was nice, but in comparison to other nicer restaurants I've been to, I didn't think this one was worth the price tag.

Table 4
Basic linguistic features.

Feature	Description
syllable	Total number of syllables
lexicon	Total number of lexicons
sentencet	Total number of sentences
char	Total number of chars
letter	Total number of letters
polysyllab	Total number of polysyllabs
monosyllab	Total number of monosyllabs
nouns	Total number of nouns
adj	Total number of adjectives
verbs	Total number of verbs
pronoun	Total number of pronouns
adverb	Total number of adverbs
article	Total number of articles

choices can serve as subtle indicators of manipulation. To capture such patterns, we construct a set of basic linguistic features designed to differentiate AI-generated fake reviews from genuine human-written ones. These features include metrics related to lexical composition, syntactic structure, and word usage frequency. A detailed list, along with their descriptions, is provided in [Table 4](#).

3.1.2. LIWC-based features

LIWC is a widely adopted tool in text analysis, offering a rich array of linguistic, cognitive, and psychological features [40]. Prior studies on fake review detection have leveraged LIWC-based features as key indicators due to their ability to capture nuanced language patterns associated with deceptive content [10]. In line with this, our study incorporates a series of LIWC-derived linguistic features to enhance the detection of AI-generated fake reviews. These features provide valuable insights into the psychological and emotional dimensions of review texts, which may differ between authentic and AI-generated content. [Table 5](#) presents the specific LIWC features utilized in this study, along with their corresponding descriptions.

3.1.3. Readability features

Readability features are also valuable indicators for distinguishing AI-generated fake reviews from genuine reviews written by humans. Since human-written or low-level machine-generated fake reviews are typically less readable than authentic reviews, much of the prior literature has employed text readability as a key feature in fake review detection [41]. However, the emergence of LLMs has reversed this

Table 5
LIWC features.

Feature	Description
space	Total number of space words
motion	Total number of motion words
cause	Total number of cause words
feel	Total number of feel words
future	Total number of future words
past	Total number of past words
present	Total number of present words
insight	Total number of insight words
tentative	Total number of tentative words

Table 6
Readability features.

Feature	Description
SMOG_Index	SMOG Index score
Flesch_Reading_Ease	Flesch Reading Ease Score
Flesch_Kincaid_Grade	Flesch Kincaid Grade score
Fog_Scale	Fog Scale score
Dale_Chall_Readability	Dale-Chall Readability Score
Reading_Time	Reading time

Table 7
Emotional lexical features.

Feature	Description
sentiment	Sentiment score
subjectivity	Subjectivity score
anger	Total anger words
sadness	Total sadness words
posemo	Total number of positive emotion words
negate	Total number of negation words
anx	Total number of anxiety words

trend, i.e., AI-generated fake reviews now tend to be more readable than those written by humans. This shift motivates the use of readability features to detect AI-generated fake reviews, particularly those with readability scores significantly higher than human-written genuine reviews. The specific readability features and their descriptions are listed in [Table 6](#).

3.1.4. Emotional lexicon features

Fake reviews generated by merchants using AI often exhibit extreme emotional polarity, particularly when the intent is to artificially enhance the ratings of their own products or services or to damage the reputation of competitors. For example, Luca and Zervas empirically demonstrated that fake reviews tend to have more extreme ratings than genuine ones [14]. Based on this insight, we construct a set of features derived from emotional vocabulary to detect AI-generated fake reviews. The emotional lexical features and their descriptions are presented in [Table 7](#).

3.2. Novel linguistic features

The traditional linguistic features are effective in detecting certain types of AI-generated fake reviews, particularly those created by merchants to manipulate product or service ratings. However, in real-world scenarios, consumers may also use AI tools to assist in writing honest reviews, as illustrated in [Fig. 1](#). These reviews typically do not exaggerate or disparage the product, making them difficult to detect using traditional linguistic cues. To address this challenge and ensure the robustness of the method, we incorporate two novel linguistic features, i.e., perplexity and burstiness, into our detection framework.

Table 8
Novel linguistic features.

Feature	Description
perplexity	Perplexity score of the whole review
perplexity_per_line	Perplexity score of each line
burstiness	Burstiness score

3.2.1. Perplexity

Perplexity is defined as the degree to which a LLM understands a given text. As a key metric for evaluating LLM performance [15,30], perplexity quantifies the model's uncertainty in predicting the next word in a sequence. Notably, human-written texts tend to be more unpredictable than AI-generated texts, making it more difficult for an LLM to anticipate subsequent words. As a result, human-authored content exhibits higher perplexity than AI-generated content. In this study, we construct two features based on the perplexity metric to detect AI-generated fake reviews: the overall perplexity of the entire review and the individual perplexity of each line within the review text.

3.2.2. Burstiness

Another innovative feature is burstiness, which refers to the likelihood of sudden occurrences of specific words or phrases within a text [15]. Compared to human-written reviews, AI-generated fake reviews typically exhibit lower variability in linguistic characteristics such as sentence structure and text length, resulting in reduced burstiness [42]. Similar to perplexity, burstiness has been widely adopted in methods for detecting AI-generated text.⁸ In this study, we integrate burstiness as a novel linguistic feature to enhance the detection of AI-generated fake reviews. The complete list of novel linguistic features used in this study, along with their descriptions, is provided in Table 8.

3.3. Anomaly probability calculation

The classical statistical principle posits that outliers typically reside in the tails of a distribution curve as low-probability events. Building on this foundation, we develop the AGFRDCP method as a primary contribution to both research and practice in detecting AI-generated fake reviews. While prior studies, such as Kumar et al. [9], have utilized feature distributions to identify fake reviews, a key limitation lies in their approximation of a data point's probability using probability density, which may introduce bias into the results. To address this limitation, we employ the cumulative probability density of a data point to more accurately assess its likelihood of being an outlier. Furthermore, since the distribution of most variables is highly skewed, outliers may appear on different sides of the distribution with varying probabilities [18]. Therefore, we weight the cumulative probability of each data point according to the skewness of its corresponding distribution, enhancing the precision of outlier detection.

In Step 1, we transform the previously constructed two categories of variables into their optimal distributions within a predefined distribution family using the Maximum Likelihood Estimation (MLE) method. The distribution family includes Normal, Uniform, Exponential, Beta, Gamma, Lognormal, and Power Law distributions.

In Step 2, we calculate the cumulative probability that each variable of a review data point conforms to its corresponding optimal univariate distribution. It is important to note that directly using cumulative probability to detect outliers is valid only when the data point lies on the left side of the distribution, as the underlying assumption of our method is that lower cumulative probabilities indicate a higher likelihood of being an outlier. To satisfy this assumption, we transform the cumulative probability of the k th variable for i th review data

point located on the right side of the distribution. The transformed cumulative probability is computed as follows:

$$\hat{F}_{\text{right}}(x_i^k) = 1 - F_{\text{right}}(x_i^k), \quad (1)$$

where $F_{\text{right}}(x_i^k)$ is the cumulative probability of k th variable of i th review data point. In this study, the mean of the k th variable is used as the threshold to determine whether a review data point lies on the left or right side of its corresponding distribution. Specifically, if the value of a review data point is less than the mean, it is considered to be on the left side of the distribution; otherwise, it is classified as being on the right side.

In Step 3, we apply weights to the cumulative probabilities calculated in the previous step based on the shape of the distribution, i.e., whether it is left-skewed or right-skewed. This adjustment is necessary because skewed distributions often result in differing probabilities of outlier occurrence on either side. For example, in a left-skewed distribution, data points on the left-hand side are more likely to be outliers. To reflect this, we multiply the cumulative probability of data points on the left side by a weight between 0 and 1, thereby increasing their likelihood of being identified as outliers. The weight for the k th feature is defined as follows:

$$W_k = \frac{1}{|\text{skewness}_k| + 1}, \quad (2)$$

where skewness_k is the skewness of the k th variable.

The AGFRDCP method is constructed as shown in Algorithm 1.

Algorithm 1: AGFRDCP method

Input: Review dataset $\mathbf{X} = \{x_i\}_{i=1}^n \in \mathbb{R}^{n \times K}$
Output: Outlier score $F(x_i)$

- 1 Compute the optimal univariate distribution p^k of x^k using MLE:
 $p^k \triangleq (p^1, \dots, p^k, \dots, p^K) = \text{Estimated univariate distribution of variables}$
- 2 Compute the skewness of the k -th variable's distribution:

$$\gamma_k = \frac{\frac{1}{n} \sum_{i=1}^n (x_i^k - \bar{x}^k)^3}{(\frac{1}{n} \sum_{i=1}^n (x_i^k - \bar{x}^k)^2)^{3/2}}$$
- 3 **for each feature of review** x_i^k **do**
- 4 **if** $\gamma_k \geq 0$ **then**
- 5 **if** $x_i^k < \bar{x}^k$ **then**
- 6 $F(x_i^k) = P(X \leq x_i^k)$;
- 7 **else**
- 8 $F(x_i^k) = \frac{1}{\gamma_k + 1} \times (1 - P(X \leq x_i^k))$ using Eq.(1) and Eq.(2)
- 9 **end**
- 10 **else**
- 11 **if** $x_i^k < \bar{x}^k$ **then**
- 12 $F(x_i^k) = \frac{1}{|\gamma_k| + 1} \times P(X \leq x_i^k)$ using Eq.(2)
- 13 **else**
- 14 $F(x_i^k) = 1 - P(X \leq x_i^k)$ using Eq.(1)
- 15 **end**
- 16 **end**
- 17 **end**
- 18 **Return** Outlier score $F(x_i) \triangleq (F(x_i^1), \dots, F(x_i^k), \dots, F(x_i^K))$

3.4. Classification models

Our AI-generated fake review detection framework employs the AdaBoost classification model. Specifically, AdaBoost is an ensemble learning method that iteratively trains multiple weak classifiers on the training data and combines them to form a strong classifier, often achieving superior performance compared to individual machine learning models [43,44]. The prediction process of AdaBoost operates through a weighted voting mechanism among its constituent weak learners. This structure allows for the examination of each weak learner's contribution, thereby enabling users to understand how the model arrives at specific decisions. As a result, the AdaBoost ensemble model offers inherent interpretability and supports both global and local interpretability, distinguishing it from deep learning models that typically function as black boxes.

⁸ Please see <https://gptzero.me/>.

4. Evaluation

4.1. Datasets and features

In this study, three datasets are employed to evaluate the performance of the proposed method. The GPT-3 validation dataset comprises 12,003 AI-generated fake reviews and 11,997 human-written reviews, where human-written reviews are verified by Yelp.com, and fake reviews are generated by GPT-3 based on these authentic samples [33]. The GPT-2 validation dataset includes 20,216 genuine reviews from Amazon and 20,215 fake reviews generated by a fine-tuned GPT-2 model. Additionally, the Llama 2 validation dataset is manually constructed for this study, consisting of 10,000 human-written reviews verified by Yelp and 10,000 fake reviews generated by Llama 2.⁹ Specifically, human-written reviews are randomly selected from the open dataset created by Gambetti et al. [33], and AI-generated reviews are produced by prompting Llama 2 with: “Write a review based on this review: [example of elite review text].” Overall, fake reviews are generated by different LLMs, while human-written reviews are sourced from two distinct e-commerce platforms. The diverse range of datasets enhances the robustness and generalizability of the proposed detection method.

4.2. Evaluation measures

Following established methodologies in the literature [10,45], we employ four evaluation metrics, i.e., accuracy, precision, recall and F-score, to assess the effectiveness of the proposed approach. Specifically, accuracy denotes the proportion of correctly identified reviews (both authentic human-written and AI-generated deceptive reviews) relative to the total number of reviews. Precision refers to the proportion of correctly identified AI-generated fake reviews among all reviews classified as such. Conversely, recall measures the proportion of correctly identified AI-generated fake reviews relative to the total number of actual AI-generated fake reviews. Lastly, the F-score represents the harmonic mean of precision and recall, providing a balanced assessment of both metrics.

4.3. Model tasks and baselines

Task 1 involves conducting a series of ablation experiments and statistical tests to examine the impact of the Traditional Linguistic Variable set (TLV) and the Novel Linguistic Variable set (NLV) on the detection performance of the AGFRDCP model. In addition to the AdaBoost algorithm, we incorporate several widely recognized machine learning classifiers [46], namely, K-Nearest Neighbors (KNN) [9], Logistic Regression (LR) [9,47], Multilayer Perceptron (MLP) [6,48], and Support Vector Machine (SVM) [49,50], to comprehensively evaluate the individual and combined contributions of TLV and NLV to the model’s detection capabilities.

Task 2 aims to demonstrate the advantages of the CPM by benchmarking it against three types of baseline methods. The CPM identifies outliers by computing the cumulative probability of each data point, thereby mitigating the bias introduced when probability density values are used in place of actual probabilities. As a first baseline, we directly apply probability density values for outlier detection to validate the effectiveness of CPM’s probabilistic approach. Second, we compare CPM with a method recently proposed by Kumar et al. [9], which employs a log-linear transformation of feature-wise probability density values to detect fake reviewers. In our implementation, we substitute the log-linear transformation with a linear transformation to accommodate feature values that are zero. Third, we assess the contribution of CPM’s innovative weighting mechanism, which adjusts the influence of each

Table 9

The tasks and according baselines methods.

Tasks	Categories	Baseline methods
Task 1	Ablation experiment	TLV NLV
Task 2	Comparative experiment	Probability Probability + LM No-weighting CPM LSTM BiLSTM
Task 3	Comparative experiment	GRU BiGRU chatgpt-detector-roberta roberta-base-openai-detector

review data point based on the shape of the underlying distribution. This weighting is hypothesized to enhance the accuracy of detecting AI-generated fake reviews. To isolate the effect of this component, we use an unweighted version of CPM as a benchmark. The same set of classification models employed in Task 1 is utilized here to ensure consistency and to illustrate the comparative advantages of CPM.

Task 3 entails a comparative evaluation of the AGFRDCP model against several state-of-the-art AI-generated fake review detection methods based on deep learning and LLMs. Specifically, we adopt a suite of advanced deep learning architectures as baseline models [51], including Long Short-Term Memory (LSTM), Bidirectional LSTM (BiLSTM), Gated Recurrent Unit (GRU), and Bidirectional GRU (BiGRU). Consistent with prior literature, these models are trained using word embeddings derived from the Word2Vec model, rather than the numerical feature representations constructed in this study. In addition, we incorporate two fine-tuned RoBERTa-based LLM detectors [30,52], shown in Table 9, as benchmark methods to further validate the superiority of AGFRDCP. These models serve as strong baselines due to their demonstrated effectiveness in detecting AI-generated content.

5. Result

5.1. Task 1

Task 1 involves conducting an ablation study to assess the contribution of the NLV to the detection performance of AGFRDCP. Specifically, we evaluate the model using three feature configurations: hybrid variable set (TLV + NLV), TLV, and NLV, across three benchmark datasets. The results, presented in Tables 10, 11, and 12, demonstrate that the hybrid feature set consistently yields superior detection performance compared to either TLV or NLV alone. This suggests that the integration of both traditional and novel linguistic cues enables the model to capture a broader and more nuanced range of suspicious patterns associated with AI-generated fake reviews.

Notably, the findings also indicate that NLV alone contributes more significantly to detection performance than TLV, underscoring the value of incorporating novel linguistic features in this context. To further validate the impact of NLV, we conduct a repeated-measures ANOVA, treating NLV as a within-subjects factor (binary: included vs. not included) and each performance metric as a dependent variable. The results, summarized in Tables 13, 14, and 15, reveal that the inclusion of NLV has a statistically significant positive effect on the performance of AGFRDCP across all datasets ($p < 0.001$).

5.2. Task 2

Task 2 presents a comparative analysis of CPM against several baseline methods across three datasets. The performance results, summarized in Tables 16, 17, and 18, demonstrate that CPM consistently outperforms the baseline approaches in terms of accuracy, precision, recall and F-score. Although CPM underperforms relative to the method

⁹ Please see <https://ai.meta.com/llama/>

Table 10
Results of ablation experiment (AI-generated review dataset).

Classifier	Feature set	Accuracy	Precision	Recall	F-score
KNN	TLV	0.7118	0.7335	0.6658	0.698
	NLV	0.77	0.7953	0.7274	0.7598
	TLV+ NLV	0.7947	0.8377	0.7311	0.7808
LR	TLV	0.76	0.7783	0.7274	0.752
	NLV	0.7878	0.771	0.8191	0.7943
	TLV+ NLV	0.8362	0.8632	0.7991	0.8299
MLP	TLV	0.7693	0.8066	0.7088	0.7545
	NLV	0.8018	0.8578	0.7238	0.7851
	TLV+ NLV	0.8383	0.8996	0.7617	0.825
SVM	TLV	0.7762	0.8216	0.7058	0.7593
	NLV	0.8033	0.8528	0.7334	0.7886
	TLV+ NLV	0.8443	0.9136	0.7607	0.8302
AdaBoost	TLV	0.7688	0.7754	0.7571	0.7661
	NLV	0.8023	0.8586	0.7241	0.7856
	TLV+ NLV	0.8387	0.8642	0.8037	0.8329

Table 11
Results of ablation experiment (GPT2-generated review dataset).

Classifier	Feature set	Accuracy	Precision	Recall	F-score
KNN	TLV	0.7611	0.7526	0.7778	0.765
	NLV	0.8123	0.8234	0.7952	0.8091
	TLV+ NLV	0.8233	0.8061	0.8514	0.8281
LR	TLV	0.7917	0.7941	0.7877	0.7909
	NLV	0.7073	0.6699	0.8174	0.7363
	TLV+ NLV	0.8548	0.8491	0.8629	0.8559
MLP	TLV	0.8219	0.8344	0.8033	0.8185
	NLV	0.8258	0.8665	0.7703	0.8155
	TLV+ NLV	0.8776	0.8688	0.8896	0.8791
SVM	TLV	0.8307	0.8235	0.8419	0.8326
	NLV	0.8239	0.841	0.7988	0.8194
	TLV+ NLV	0.902	0.891	0.9159	0.9033
AdaBoost	TLV	0.8152	0.817	0.8124	0.8147
	NLV	0.8221	0.8751	0.7515	0.8086
	TLV+ NLV	0.9092	0.921	0.8951	0.9079

Table 12
Results of ablation experiment (Meta-generated review dataset).

Classifier	Feature set	Accuracy	Precision	Recall	F-score
KNN	TLV	0.9064	0.8785	0.9432	0.9097
	NLV	0.9302	0.927	0.934	0.9305
	TLV+ NLV	0.9642	0.95	0.98	0.9648
LR	TLV	0.912	0.9011	0.9256	0.9132
	NLV	0.9272	0.9124	0.9452	0.9285
	TLV+ NLV	0.9764	0.9734	0.9796	0.9765
MLP	TLV	0.9114	0.8997	0.926	0.9127
	NLV	0.9288	0.9298	0.9276	0.9287
	TLV+ NLV	0.9768	0.9749	0.9788	0.9768
SVM	TLV	0.932	0.9239	0.9416	0.9326
	NLV	0.9318	0.9268	0.9376	0.9322
	TLV+ NLV	0.9824	0.9805	0.9844	0.9824
AdaBoost	TLV	0.9424	0.9378	0.9476	0.9427
	NLV	0.9354	0.9297	0.942	0.9358
	TLV+ NLV	0.9836	0.9828	0.9844	0.9836

Table 13
Results of statistical testing (AI-generated review dataset).

Classifier	Accuracy		Precision		Recall		F-score	
	F-value	P-value	F-value	P-value	F-value	P-value	F-value	P-value
KNN	1443.7	<0.001	1569.8	<0.001	222.1	<0.001	929.5	<0.001
LR	1599	<0.001	1108.6	<0.001	358.6	<0.001	1251.2	<0.001
MLP	673.8	<0.001	384.8	<0.001	31.2	<0.001	375.4	<0.001
LR	887.6	<0.001	957.1	<0.001	136.8	<0.001	665.5	<0.001
AdaBoost	957.8	<0.001	887.2	<0.001	117.5	<0.001	756.5	<0.001

Table 14
Results of statistical testing (GPT2-generated review dataset).

Classifier	Accuracy		Precision		Recall		F-score	
	F-value	P-value	F-value	P-value	F-value	P-value	F-value	P-value
KNN	1941.2	<0.001	875.6	<0.001	2317	<0.001	2475.2	<0.001
LR	4483	<0.001	1421.5	<0.001	1724.7	<0.001	4565.8	<0.001
MLP	313.9	<0.001	143.4	<0.001	114.1	<0.001	305.8	<0.001
LR	3404.7	<0.001	1915.2	<0.001	1845.2	<0.001	3395.2	<0.001
AdaBoost	13 057.8	<0.001	5939.2	<0.001	1339	<0.001	9798	<0.001

Table 15
Results of statistical testing (Meta-generated review dataset).

Classifier	Accuracy		Precision		Recall		F-score	
	F-value	P-value	F-value	P-value	F-value	P-value	F-value	P-value
KNN	2026.1	<0.001	1765.7	<0.001	832.8	<0.001	2025.6	<0.001
LR	3193.9	<0.001	2216.7	<0.001	929.9	<0.001	3120.7	<0.001
MLP	2535.5	<0.001	1085.7	<0.001	350	<0.001	2458.2	<0.001
LR	1789.6	<0.001	1565.1	<0.001	761.2	<0.001	1795	<0.001
AdaBoost	1794.5	<0.001	1172.1	<0.001	914.4	<0.001	1801.9	<0.001

Table 16
Results of Task 2 (AI-generated review dataset).

Classifier	Method	Accuracy	Precision	Recall	F-score
KNN	Probability	0.6078	0.6461	0.4775	0.5491
	Probability + LM	0.63	0.6377	0.6025	0.6196
	No-weighting	0.7185	0.7105	0.7378	0.7239
	AGFRDCP	0.7947	0.8377	0.7311	0.7808
LR	Probability	0.6078	0.6461	0.4775	0.5491
	Probability + LM	0.63	0.6377	0.6025	0.6196
	No-weighting	0.7185	0.7105	0.7378	0.7239
	AGFRDCP	0.7947	0.8377	0.7311	0.7808
MLP	Probability	0.5015	0.5008	0.9847	0.664
	Probability + LM	0.5023	0.5013	1	0.6678
	No-weighting	0.7633	0.7598	0.7704	0.7651
	AGFRDCP	0.8383	0.8996	0.7617	0.825
SVM	Probability	0.5953	0.6134	0.5165	0.5608
	Probability + LM	0.6108	0.5788	0.8154	0.677
	No-weighting	0.7928	0.826	0.7421	0.7818
	AGFRDCP	0.8443	0.9136	0.7607	0.8302
AdaBoost	Probability	0.8307	0.8443	0.8111	0.8273
	Probability + LM	0.8373	0.8564	0.8107	0.8329
	No-weighting	0.7825	0.7908	0.7684	0.7794
	AGFRDCP	0.8387	0.8642	0.8037	0.8329

proposed by Kumar et al. [9] (Probability + LM) on the Llama 2 validation dataset, it achieves significantly superior results on the other two datasets, highlighting its overall robustness and effectiveness. Furthermore, the findings reveal that the direct use of cumulative probabilities for outlier detection does not inherently guarantee superior performance over probability density-based methods, particularly when the distributional shape of outliers is not accounted for. While cumulative probabilities address certain limitations of density-based approaches, neglecting the influence of distributional shape can introduce bias into the detection process. This underscores the importance of CPM's design, which incorporates distribution-aware weighting to enhance the reliability of AI-generated fake review detection.

5.3. Task 3

Task 3 presents a comparative performance analysis of AGFRDCP against baseline methods based on deep learning and LLMs across three benchmark datasets. The results, detailed in Tables 19, 20, and 21, demonstrate that AGFRDCP consistently outperforms all baseline models in terms of accuracy, precision and F-score. These empirical findings underscore the effectiveness of the proposed method in accurately identifying AI-generated fake reviews. Moreover, the superior performance

Table 17
Results of Task 2 (GPT2-generated review dataset).

Classifier	Method	Accuracy	Precision	Recall	F-score
KNN	Probability	0.5814	0.5657	0.7008	0.6261
	Probability + LM	0.6172	0.6181	0.6134	0.6158
	No-weighting	0.7725	0.7426	0.834	0.7856
	AGFRDCP	0.8233	0.8061	0.8514	0.8281
LR	Probability	0.569	0.5627	0.6187	0.5894
	Probability + LM	0.6007	0.5801	0.7293	0.6462
	No-weighting	0.7142	0.7083	0.7283	0.7182
	AGFRDCP	0.8548	0.8491	0.8629	0.8559
MLP	Probability	0.5262	0.5148	0.9141	0.6586
	Probability + LM	0.5326	0.5174	0.9683	0.6745
	No-weighting	0.8545	0.8447	0.8686	0.8565
	AGFRDCP	0.8776	0.8688	0.8896	0.8791
SVM	Probability	0.5634	0.5595	0.5964	0.5773
	Probability + LM	0.5631	0.5592	0.5964	0.5772
	No-weighting	0.8389	0.8195	0.8694	0.8437
	AGFRDCP	0.902	0.891	0.9159	0.9033
AdaBoost	Probability	0.881	0.8827	0.8787	0.8807
	Probability + LM	0.8741	0.8737	0.8746	0.8741
	No-weighting	0.864	0.87	0.8558	0.8628
	AGFRDCP	0.9079	0.9198	0.8937	0.9066

Table 18
Results of Task 2 (Meta-generated review dataset).

Classifier	Method	Accuracy	Precision	Recall	F-score
KNN	Probability	0.583	0.5857	0.5672	0.5763
	Probability + LM	0.5446	0.5475	0.5144	0.5304
	No-weighting	0.77	0.7339	0.8472	0.7865
	AGFRDCP	0.9642	0.95	0.98	0.9648
LR	Probability	0.557	0.5454	0.6844	0.6071
	Probability + LM	0.5678	0.5944	0.4268	0.4969
	No-weighting	0.7616	0.7565	0.7716	0.764
	AGFRDCP	0.9764	0.9734	0.9796	0.9765
MLP	Probability	0.5022	0.5011	0.9992	0.6675
	Probability + LM	0.5078	0.5397	0.106	0.1772
	No-weighting	0.9076	0.9126	0.9016	0.907
	AGFRDCP	0.9768	0.9749	0.9788	0.9768
SVM	Probability	0.5348	0.5303	0.6084	0.5667
	Probability + LM	0.5352	0.5312	0.5984	0.5628
	No-weighting	0.9034	0.9068	0.8992	0.903
	AGFRDCP	0.9824	0.9805	0.9844	0.9824
AdaBoost	Probability	0.974	0.9717	0.9764	0.9741
	Probability + LM	0.9852	0.9856	0.9848	0.9852
	No-weighting	0.9186	0.9184	0.9188	0.9186
	AGFRDCP	0.9834	0.9828	0.984	0.9834

Table 19
Results of Task 3 (AI-generated review dataset).

Method	Accuracy	Precision	Recall	F-score
LSTM	0.7981	0.8694	0.704	0.778
BiLSTM	0.8054	0.8805	0.709	0.7855
GRU	0.7973	0.8699	0.7015	0.7767
BiGRU	0.7925	0.8856	0.6741	0.7655
ChatGPT_detector	0.6916	0.8776	0.4455	0.591
RoBERTa_base model	0.7689	0.8514	0.6517	0.7383
AGFRDCP	0.8387	0.8642	0.8037	0.8329

of AGFRDCP highlights its practical potential for deployment in real-world e-commerce platforms, where robust and reliable detection of deceptive content is critical.

5.4. Explainable AGFRDCP

To enhance the interpretability and transparency of AGFRDCP, we employ the Local Interpretable Model-agnostic Explanations (LIME) framework to elucidate the model's decision-making process. LIME generates locally faithful explanations by perturbing input features, such as review text, sentiment indicators, and linguistic patterns. It also identifies the most influential features that differentiate AI-generated fake

Table 20
Results of Task 3 (GPT2-generated review dataset).

Method	Accuracy	Precision	Recall	F-score
LSTM	0.8299	0.7988	0.8787	0.8369
BiLSTM	0.8229	0.7748	0.9071	0.8357
GRU	0.8432	0.8368	0.8553	0.846
BiGRU	0.8501	0.8442	0.8612	0.8526
ChatGPT_detector	0.5364	0.6479	0.1595	0.256
RoBERTa_base model	0.8388	0.7789	0.9462	0.8544
AGFRDCP	0.9092	0.921	0.8951	0.9079

Table 21
Results of Task 3 (Meta-generated review dataset).

Method	Accuracy	Precision	Recall	F-score
LSTM	0.927	0.9311	0.9207	0.9259
BiLSTM	0.959	0.9536	0.9642	0.9588
GRU	0.9435	0.9201	0.9702	0.9445
BiGRU	0.9583	0.9472	0.9697	0.9583
ChatGPT_detector	0.8600	0.9270	0.7815	0.8481
RoBERTa_base model	0.8252	0.8690	0.7659	0.8142
AGFRDCP	0.9834	0.9828	0.984	0.9834

reviews from human-authored ones. This interpretability mechanism offers actionable insights for stakeholders, enabling them to understand the rationale behind the classification of a review as synthetic. Fig. 4 illustrates a representative explanation from the GPT-3 validation dataset, where the review was classified as an “AI-generated fake review”. Among the features analyzed, the perplexity of the entire review emerged as the most influential predictor in the model's decision. This level of transparency not only supports the credibility of the detection process but also fosters greater trust among e-commerce platforms and end-users in the deployment of AGFRDCP.

6. Conclusion

6.1. Summary of results

Designing effective algorithms to prevent the misuse of LLMs, such as ChatGPT, for disseminating false information has become an increasingly critical and urgent challenge for e-commerce platforms [26]. The scope of the proposed method is to detect AI-generated fake reviews, recognizing that online reviews play a pivotal role for consumers, merchants, and platforms within the broader platform ecosystem [53–55]. The major findings and potential alternative explanations are summarized as follows.

First, we employ the predefined feature construction method that incorporates two types of unique linguistic features and uses them as input for the classification model. Numeric results of the ablation experiments conducted in Task 1 show that AGFRDCP, when incorporating both TLV and NLV, achieves superior detection accuracy compared to using either feature set alone. This finding confirms that the TLV and NLV developed in this study effectively capture critical linguistic cues indicative of AI-generated fake reviews, thereby playing a pivotal role in enhancing AGFRDCP's performance. While much of the existing literature on AI-generated fake review detection has primarily concentrated on fine-tuning the parameters of LLMs to enhance detection performance [32,33], often overlooking the foundational linguistic features of reviews and the metrics applicable for evaluating different LLMs.

Second, we introduce an outlier detection method to calculate the probability that a fake review generated by AI. This approach addresses two key limitations in existing methods: it corrects the theoretical bias introduced by substituting probability values with probability density values [9], and it resolves the challenge of determining appropriate probability thresholds to distinguish outliers from normal observations [17,18]. Overall, theoretical analysis suggests that CPM, which leverages weighted cumulative probability densities, is better suited

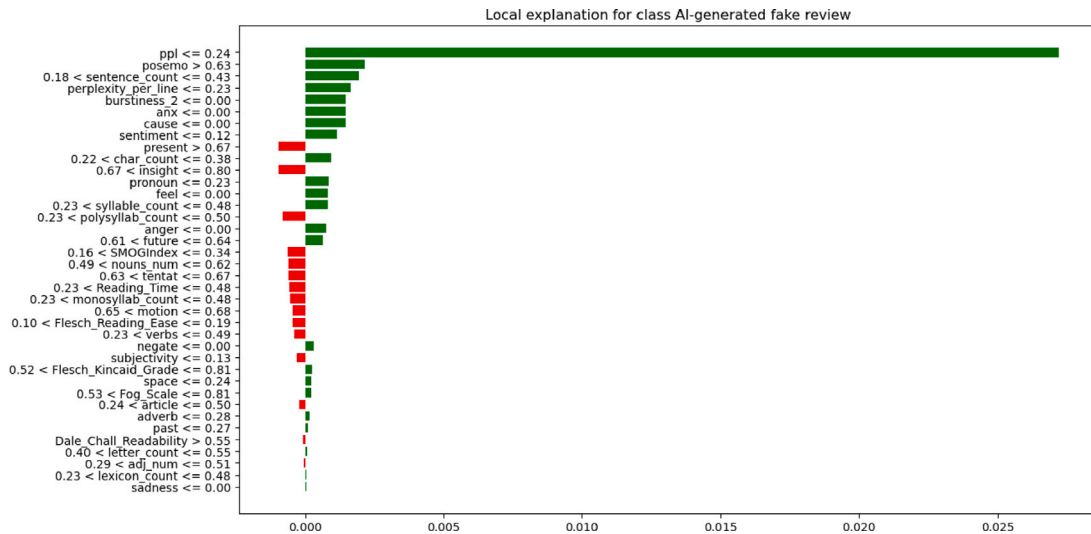


Fig. 4. LIME explanation.

for identifying anomalies than conventional baseline methods. This conclusion is empirically supported by the comparative experiments in Task 2, which validate CPM's superior outlier detection capability.

Last, we propose the AGFRDCP integrated the constructed feature sets, CPM and AdaBoost classification model to detect AI-generated fake reviews. Numeric results of Task 3 reveal that AGFRDCP outperforms baseline models based on deep learning and LLMs across multiple datasets. These results collectively underscore the effectiveness and innovation of AGFRDCP in detecting AI-generated fake reviews with high accuracy. More important, the proposed method enhances transparency in platform decision-making processes [56–58], in contrast to prior research that primarily relies on fine-tuning LLMs for detection tasks [32,33]. To the best of our knowledge, this study is among the first to introduce such a novel and interpretable framework within the literature on AI-generated fake review detection. By incorporating pre-defined feature construction method, CPM and AdaBoost classification model, this study improves the interpretability of detection method, thereby overcoming the inherent “black box” issue of deep learning-based detection methods.

6.2. Theoretical implications

First, this study addresses a critical gap in the literature concerning feature construction for detecting AI-generated fake reviews by proposing a novel detection framework that integrates both conventional linguistic features and newly introduced linguistic indicators. This comprehensive feature set facilitates the development of a more robust and nuanced set of predictors for identifying AI-generated deceptive content. To the best of our knowledge, this integrative approach is unprecedented in the existing literature. Through a series of empirical evaluations, we demonstrate that incorporating these variables significantly enhances the predictive accuracy of detection models.

Second, this study advances the literature by introducing a novel outlier detection technique based on cumulative probability. This method addresses a critical bias in prior outlier detection research, which often substitutes probability values with probability density values, thereby neglecting the classical statistical principle of cumulative probability [18]. By leveraging cumulative probability to estimate the likelihood that a review is AI-generated, the proposed method mitigates this bias. Furthermore, recognizing that the shape of the distribution influences the probability of a data point being an outlier, we introduce a weighting mechanism that adjusts cumulative probabilities accordingly. This enhancement further improves the accuracy of outlier detection. As anticipated, experimental results confirm that the

proposed method outperforms several well-established outlier detection techniques.

Last, this study contributes to the literature on AI-generated fake review detection by being among the first to introduce a machine learning-based approach specifically designed for this task. Unlike prior methods that primarily rely on fine-tuning LLMs, which often suffer from limited interpretability and transparency, the proposed method addresses these shortcomings while maintaining high detection performance [32,33]. Empirical evaluations conducted on three real-world datasets demonstrate the effectiveness of the proposed approach in accurately identifying AI-generated fake reviews. Notably, these datasets were generated using different LLMs, indicating that the method is capable of detecting a diverse range of AI-generated content, thereby underscoring its robustness and generalizability. These findings suggest that the proposed method holds significant practical value for e-commerce platforms seeking to filter or remove AI-generated reviews produced by various generative models.

In addition, this study offers two significant theoretical contributions to the information systems (IS) literature on deception detection and digital trust. First, while outlier detection techniques have substantially advanced anomaly detection across various IS domains, including suicidal ideation identification [59], financial fraud surveillance [35], and fake content recognition [8], prior research has predominantly relied on deep learning models, often at the expense of interpretability. In contrast, our study introduces a novel detection framework that employs weighted cumulative density estimation to quantify outlier likelihood, thereby addressing the interpretability limitations inherent in earlier approaches. Second, this research advances IS scholarship by developing a machine learning-based detection method specifically designed to counteract AI-generated fake reviews, thereby confronting the dark side of GAI. By uncovering deceptive behavioral patterns and enhancing detection precision, our work deepens the understanding of how advanced computational techniques can mitigate emergent technological risks and reinforce digital trust within e-commerce ecosystems.

6.3. Practical implications

From a practical standpoint, this study yields several important managerial implications for e-commerce platforms increasingly challenged by AI-generated fake reviews. First, the proposed method can be integrated into the decision support systems of such platforms to effectively identify and filter out AI-generated fake reviews [60]. Notably, while the prevailing approach in the field, i.e., “AI combatting AI”,

which involves developing detectors based on LLMs to identify LLM-generated content, has gained significant attention, our findings suggest that this strategy contributes only marginally to detection performance. Instead, evaluation metrics tailored to LLMs and traditional linguistic features demonstrate greater efficacy, offering valuable guidance for the design of detection systems. Second, platforms should prioritize these evaluation metrics and linguistic features to achieve more accurate detection outcomes with reduced computational resources and time investment. Furthermore, the proposed method is highly adaptable and generalizable, enabling platforms to rapidly tailor it to other detection contexts, such as identifying AI-generated news content. In summary, our approach offers a practical and scalable solution to mitigate the misuse of GAI technologies in public digital ecosystems.

6.4. Limitations and future directions

There are several limitations in this study that warrant consideration in future research. First, the current framework does not incorporate behavioral characteristics of reviewers into the AI-generated fake review detection model. To address this gap, subsequent studies could develop more comprehensive feature sets based on reviewer behavior (e.g., frequency of review submissions) to enhance detection capabilities. Second, the proposed approach omits visual data from product images, which may offer valuable multimodal signals for identifying AI-generated fake reviews. Future investigations could explore deep learning-based multimodal fusion techniques that integrate both textual and visual modalities to improve detection performance. Third, the methodology does not explicitly model inter-reviewer relationships, which is particularly important given the increasing use of GAI tools by coordinated spammers to manipulate product ratings. Constructing reviewer interaction networks and incorporating them into the detection framework may facilitate the identification of orchestrated inauthentic behavior, thereby strengthening the robustness of AI-generated fake review detection.

CRediT authorship contribution statement

Jiwei Luo: Writing – original draft, Methodology, Formal analysis, Data curation. **Guofang Nan:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Dahui Li:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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Jiwei Luo (jiweiluo@hainanu.edu.cn) is currently pursuing the Ph.D. degree in Business Administration at International Business School, Hainan University, China. His current research focuses on the machine learning techniques with applications and empirical studies of Generative AI. His recent paper has appeared in *Decision Support Systems* and *Tourism Management*.

Guofang Nan (gfnan@hainanu.edu.cn) is a professor of MIS at International Business School, Hainan University, China. He received the M.Sc. and Ph.D. degrees from the Institute of Systems Engineering, Tianjin University, Tianjin, China, in 2002 and 2004, respectively. Dr. Nan's current research interests include diverse areas of information systems, decision support systems, and information economics. He has published more than 50 papers in such journals as *Management Science*, *Information Systems Research*, *Journal of Management Information Systems*, *Journal of the Association for Information Systems*, and *Decision Support Systems*.

Dahui Li (dli@uccs.edu) is a professor of MIS at the University of Colorado Colorado Springs. He received his Ph.D. in MIS from Texas Tech University. His current research focuses on business-to-consumer relationships, online communities, and technology innovation. He has had papers published in the *Communications of the ACM*, *Decision Sciences*, *Decision Support Systems*, *Information & Management*, *Journal of the Association for Information Systems*, *Journal of Product Innovation Management*, and elsewhere.